Not every stable cluster is homeostatic: Stability, network order, maintenance, and control in projectible kinds

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Abstract

Homeostatic property cluster theory gave natural-kinds realism a powerful anti-essentialist form: a kind can be real when a clustered profile supports induction and explanation by accommodating our classifications to causal structure rather than by sharing an essence. In the HPC-descended debate, the homeostatic component can be made to cover several achievements that aren't the same: a stable projectible profile, network order, maintenance, and corrective control.

This paper treats projectibility as the target of kind claims and separates two counterfactuals that HPC talk can run together: what would happen if a STABILIZER were removed, and whether some process registers and answers perturbations to the profile. The central result is separability: a profile may be stable, network ordered, or maintained without yet being control-secured. That result holds whatever we call the top tier. If HOMEOSTATIC is to mark a distinctive achievement, it should name only the case where perturbation-sensitive organization preserves a specified higher-scale relation through lower-scale variation.

The Watt governor shows how lower-scale feedback can produce higher-scale matching; glycemic regulation shows how a regulated range can be constituted by self-maintenance; interjection tokens show projectible stability that no specified CONTROLLER secures; and monitored agreement production shows what a specified linguistic loop would have to provide.

The payoff is a diagnostic that keeps those achievements apart without turning HOMEOSTATIC into a loose synonym for stable or projectible.

Keywords: natural kinds; homeostatic property clusters; projectibility; homeostasis; causal control; linguistic categories

1 Introduction

A quiet ambiguity runs through recent anti-essentialist kind theory. Boyd's homeostatic property cluster framework, Slater's stable property cluster proposal, and Khalidi's causal-network account all

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reject essences, and they all aim to explain why some categories support projection while others do not. Each, though, locates the basis for projection somewhere different.

Boyd ties projectibility to homeostatic mechanisms that maintain the cluster (Boyd, 1991); Slater treats natural kindness as a matter of cliquish stability rather than of any particular maintaining mechanism (Slater, 2015); and Khalidi locates projectibility in the causal-network relations among core and derivative properties (Khalidi, 2013, 2018). In this journal, Onishi and Serpico (2022) argue that the notion of a homeostatic mechanism in HPC theory remains underexplicated. The present paper takes that opening seriously. The title names a pressure inside that vocabulary: even when no one explicitly identifies every stable cluster with a homeostatic one, HPC-descended vocabulary can leave the securing structure for a given projection underdetermined.

The looseness is clearest in the homeostatic-mechanism component of the framework. Boyd gives that component considerable flexibility, treating the clustering as "metaphorically (sometimes literally)" homeostatic and allowing it to be sustained by mutual property support, underlying mechanisms, or both (Boyd, 1999). That flexibility is a virtue for anti-essentialist realism, but it also makes it easy to slide from stable profile to stabilizer, from stabilizer to homeostasis, and from homeostasis to projectible kind, as though these were one achievement.

The order should run the other way. Start from a projection target, ask what empirical profile supports it, ask what ordering relations make the profile non-accidental, ask what maintains it, and only then ask whether the maintenance is corrective control. This order keeps Boyd's projectibility frame in place while asking that HOMEOSTASIS be used in a more demanding, control-theoretic sense. Weinberger's recent account of homeostasis as causal control supplies the strictest standard (Weinberger, 2026); the contribution here is to place that standard inside an HPC-facing taxonomy of weaker securing structures.

The answer to Onishi and Serpico (2022) is therefore a division of evidential labour. The original work of the paper is the STABILIZER/CONTROLLER split and the intermediate taxonomy around it: a maintainer can stabilize a projectible profile without registering and correcting perturbations to the relation that made it projectible. Network order and maintenance are likewise distinct evidential demands, and many real kinds may satisfy some of these demands without satisfying others.

The cases have to show that separability. The scope is Boyd-style projectible realist kind claims. Many classifications earn their keep by mapping variation, stabilizing measurement, organizing historical comparison, or coordinating inquiry before they support strong projection. Those uses are legitimate, but the diagnostic here concerns categories offered as real kinds because they support prediction, explanation, induction, or controlled intervention.

The paper tests the diagnostic across a machine, an organism, and two linguistic phenomena. The linguistic payoff is sober: the diagnostic mostly tells us when not to call a category HOMEOSTATIC, while preserving a standard for future positive cases.

2 Projectibility first

Boyd's framework works because projectibility doesn't require necessary and sufficient conditions. A category can support induction when its associated properties are connected non-accidentally, even where no single property is shared by every member. On Boyd's accommodation thesis, successful classification is a matter of fitting our inferential practices to causal structure, so that induction and explanation "always require that we accommodate our categories to the causal structure of the world"

(Boyd, 1991, p. 224). We construct the categories, but whether they succeed depends on a structure we don't control. That structure carries the explanation.

The problem of PROJECTIBILITY comes from Goodman's new riddle of induction, where a predicate can fit all observed cases and still fail to license projection to unobserved ones (Goodman, 1955/1983).

Goodman sets up the problem here; the diagnostic departs from his entrenchment response. That response treats projectibility as inherited from a predicate's history of successful projection. The present diagnostic follows Boyd in asking a different question: once a projection target is declared, what structure makes the projection reliable, and what empirical tests constrain the declaration against gerrymandering?

Jantzen (2015) makes the same problem explicit for scientific practice by distinguishing system induction from kind induction; the present argument uses PROJECTION TARGET broadly enough to cover both.

Boyd's contribution was to ground projectibility in causal structure without grounding it in essences, which is also what gave HPC theory its purchase against the worry, which Boyd associated with Hacking, that where a kind's boundaries are fuzzy its membership is fixed by "people in concert" rather than by nature (Boyd, 1991).

The same point can be put in Boyd's later terms. A programmatic definition states the inductive or explanatory role a kind term is supposed to play; an explanatory definition states the properties and causal structures in virtue of which it plays that role (Boyd, 1999). The projection target in the present diagnostic is programmatic in that sense. Stable profile, network order, maintenance, and control are candidate explanatory supports.

One reason, then, is that it helps to begin from the projection target before mechanism. Mechanisms matter to the theory because they explain why projection is reliable, so a category that supports no projection isn't redeemed by having a tidy causal story behind it, and a category that does support projection is already doing much of the work a realist about kinds cares about. The task of the rest of the paper is to ask, for categories that support projection, which further evidential demands they meet.

The risk is that this question can be made too easy. If the only thing asked is whether a cluster recurs, a great many categories will qualify, because recurrence on its own is cheap. Stable co-occurrence may license some inference, but an analysis still has to specify which projection is being made before the category becomes theoretically interesting, and that happens only when its profile supports prediction, explanation, or controlled intervention.

Projection is therefore field-relative without being arbitrary. A category is projectible for some family of questions, contrasts, and evidential practices: metabolic physiology, interactional linguistics, and grammar license different expectations from different profiles. But the field doesn't make the projection successful. Once the target is declared, the profile still has to support reliable expectations under the relevant contexts and interventions.

Nor is the declaration self-certifying. It is a proposal about what the profile lets us infer, and it earns its role only when the declared projection survives empirical tests under the relevant contrasts. An analyst can redescribe a profile in many ways, but the redescription won't secure a kind claim unless it improves projection.

3 Observed stability and invariance

Slater's stable property cluster account presses the useful point that natural kindness needn't be assessed by first identifying a homeostatic mechanism. What matters on this view is cliquish stability: observing enough of a cluster licenses expectations about the larger cluster across the counterfactual range relevant to a domain of inquiry.

Slater's point is that natural kindness can be assessed through this projectibility-shaped stability without first identifying the mechanism, if any, by which the stability is maintained (Slater, 2015). This yields the first distinction the paper needs, because a stable and projectible cluster may be perfectly real without being homeostatic in any stronger sense.

Once the target is fixed, stable-profile evidence is the first demand. It asks whether the profile supports the declared projection: whether observing part of the cluster licenses expectations about unobserved features or behavior across relevant contexts. That evidence can make the category real and projectible. It still leaves open which stronger securing claim has been earned. Stable-profile evidence alone doesn't show network order, identify a stabilizer, or exhibit corrective control.

The dynamical-kinds account in Jantzen (2015) is a discovery-oriented complement: it asks how projectible systems or kinds can be found by looking for dynamical symmetries, transformations to which a causal structure is indifferent. The present diagnostic asks what has been secured once a profile is proposed. Symmetry or invariance evidence may show stable profile or network order, but not maintenance or perturbation-sensitive control.

This is where Onishi and Serpico's dilemma enters. If an account stops at stable-profile evidence, it may be too slender for some explanatory projects. It can say that the projection is supported across the relevant range, but it may not explain why newly discovered properties should be expected in other members, or how the same kind supports projection across context-dependent realizations.

Craver's strict-mechanism reading shows why the problem isn't solved by demanding mechanisms: deciding which factors belong to a mechanism, when mechanisms share a type, and where one ends introduces conventional perspective into mechanism-based kind claims (Craver, 2009).

Onishi and Serpico generalize the same pressure to causal-network order. If causal structure is retained through causal-network order, as Khalidi proposes, causal graphs still require choices about boundaries, abstraction level, and where a network should be carved when processes and clusters fail to map one-to-one (Onishi & Serpico, 2022). Their broader observation is that the notion of a homeostatic mechanism has never been fully explicated, and that interest-relativity is unlikely to be eliminable.

The diagnostic offered here doesn't try to remove that interest-relativity so much as to make it explicit. Asking first what a category lets us project, and against which declared expectation, fixes the explanatory target, and the question of securing structure is then asked relative to that target rather than in the abstract. Interest-relativity becomes a parameter the analysis declares rather than a cost it pays without noticing. The purpose is human; whether the profile supports projection for that purpose is an empirical constraint.

Woodward's account of invariance adds a stronger demand for explanatory projections or projections that support intervention. Invariance is not de facto persistence across space and time. It is a modal relation: a generalization is explanatory when the relation it describes would continue to hold under an appropriate range of interventions on variables in that relation (Woodward, 2001).

The common-cause lesson matters here. A correlation may be stable across many ordinary contexts and still fail to explain if an intervention on one effect doesn't manipulate the other. The right intervention reveals counterfactual dependence; the wrong one may break, bypass, or mislocate the relation being tested.

The first diagnostic step after the projection target is correspondingly modest. A stable property cluster should support a Slater-style projection across a specified range of contexts, with evidence that goes beyond recurrence and shows that observing part of the profile licenses a useful expectation about another part. When the claimed payoff is explanation or controlled intervention, the stronger Woodwardian test asks whether the relation that licenses the projection is invariant under relevant what-if changes. That is not yet a control claim: invariance is the test; corrective control is a later claim about why the invariance holds.

4 Network order without control

Khalidi's network account addresses a different shortcoming. It also starts from projectibility: natural kind predicates support reliable inferences to other predicates, and causal-network order explains why those inferences are not accidental. A cluster is more than a list of associated properties because its properties can be ordered, with some core causal properties giving rise to derivative properties in recurrent causal processes (Khalidi, 2013, 2018).

That matters because projectibility has two dimensions in Khalidi's discussion: the variety of projections a kind supports and the generality or strictness of those projections. Network order supplies the ontological correlate. Richer and tighter causal relations explain richer and tighter projections, while a merely associated bundle of features has no comparable structure.

That order is the strength of the account. The present point is narrower: ordered dependence, even when objective and explanatory, doesn't by itself amount to feedback-sensitive correction.

Ordered dependence is still some distance from corrective control. A node in a causal network can support projection because the properties around it stand in systematic causal relations. This remains compatible with the system never tracking a departure, never allowing lower-scale variables to adjust, and never preserving a higher-scale relation through any such adjustment. Network order can explain projectibility without amounting to homeostasis.

The network-order step therefore asks whether the profile has a structure that explains which projections are licensed. The relevant evidence consists of asymmetries among the properties: which features ground or condition others, which are derivative, which are central because many projections flow through them, and which are only peripheral cues. The characteristic failure is the use of causal vocabulary for a merely unordered list, since redescribing co-occurrence in causal terms doesn't by itself establish that any ordering obtains.

I use NETWORK ORDER broadly across the diagnostic. This is an extension of Khalidi's evidential role, not a claim that every ordering relation is Khalidi-style causal naturalness. In strictly biological or mechanical cases the ordering will usually be causal, whereas in linguistic and social cases the relevant dependencies may instead be functional, conventional, sequential, normative, or evidential. What matters at this diagnostic step is that the profile has a structure that explains the declared projection.

The broad use still needs a constraint. A non-causal ordering relation should be asymmetric: one feature conditions, licenses, constrains, or structures another. Network order requires more than evidential usefulness. A cue is evidentially privileged only when it tracks an ordering relation in the profile rather than merely co-occurring with the rest of the bundle. If the analyst wants the stronger

Khalidi-style natural-kind claim, convention, evidential usefulness, or blueprint-like co-instantiation is insufficient on its own; causal, causal-historical, or causal-normative support has to be shown.

5 From stabilizers to controllers

Maintenance introduces two questions that are easy to run together. Boyd's HOMEOSTATIC label was deliberately broad: a cluster may be sustained because some properties favor others, because underlying mechanisms maintain their co-occurrence, or both (Boyd, 1999). That breadth is part of what made the framework useful.

For the diagnostic here, whatever keeps a cluster clustered can be called its STABILIZER, in the sense that removal would let the category fragment, drift, or dissolve. The stabilizer is a functional role, and the causal structure that fills the role is the mechanism. Maintenance, in this sense, is the umbrella term.

The question the paper presses is which stabilizers are also controllers. A CONTROLLER is the narrower case: a stabilizer with feedback-sensitive corrective organization that preserves a higher-scale relation through lower-scale variation. Not every stabilizer meets this condition. Reliable co-occurrence can be maintained by entrenchment, by transmission, or by a frozen accident that nothing actively corrects, and ordered dependence can be maintained by causal wiring that carries no error signal and returns the system to no target.

The two ideas answer different counterfactuals, which is why they come apart. The stabilizer question is about the maintainer, and it asks what would happen to the cluster if the maintaining process were removed. The controller question is about the profile under threat, and it asks whether some maintaining process registers and answers a perturbation that threatens the profile in a way that brings the profile back toward a relation worth projecting from.

A case can satisfy the first condition and fail the second, since entrenchment and transmission may hold a profile together without detecting or correcting a departure. The space between a merely held profile and an actively corrected one is where HOMEOSTATIC tends to be overclaimed.

Read this way, maintenance is independent of network order in both directions. A stable profile can support projection before its maintainer is known, and a network-ordered profile can explain why projection is reliable without yet showing what keeps the network in place. Conversely, a maintainer may keep a profile in place without giving the profile much internal order. Once a maintainer is identified, the further question is whether that maintainer is a controller. Only the controller variety earns HOMEOSTATIC in the strict control-theoretic sense, and the next section says what control adds.

6 Corrective control

Control-secured kind claims are more demanding, and the first thing to get right is the level at which the control is located. A controller can of course be a kind of system, but the kind label leaves open what senses, compares, or corrects. A candidate kind earns the HOMEOSTATIC label only when its projectible profile is maintained by a specified system or practice with feedback-sensitive corrective organization.

That specification is itself relative to the projection target. The analyst has to say which higher-scale relation matters, which lower-scale variables may vary, which perturbations threaten the relation, and which response pathway preserves it. The scale split therefore remains interest-relative, but it has to be declared before the case is classified.

The proposed target has to appear in the corrective dynamics: the relevant perturbation should disturb the candidate relation, the response pathway should answer that perturbation, and the preserved relation should be the one that supports the projection. Relabelling a maintained profile as control doesn't supply that pattern.

Weinberger's central lesson is that homeostasis requires more than constancy, because a controlled system can preserve a higher-scale relation precisely by letting its lower-scale variables move. His dynamic causal models show that feedback systems, and the Watt governor in particular, produce control relations that static causal models and a simple vocabulary of intervention tend to obscure (Weinberger, 2026).

In the governor case, the neglected relation is matching steam admission to workload, not holding shaft speed fixed. That point fits a broader control literature on organized systems. In Bich and Bechtel's terms, control mechanisms measure variables and use those measurements to set constraints in controllers, which then act on flexible constraints in the mechanisms they control (Bich & Bechtel, 2022a). In living systems, control is often organized across networks of mutually dependent processes rather than issued from a single command point (Bich & Bechtel, 2022b; Bich et al., 2016).

Two consequences are worth drawing out. The first is that holding a variable rigidly fixed can be a sign of less control rather than more, since appropriate variation in the lower-scale variables is often the means by which the higher-scale relation is kept in place.

The second is that not every intervention is equally probative. The intervention has to test the relation through the relevant pathway rather than destroy or bypass it. Perturbations to workload or demand can reveal the controller's response, while clamping the regulated variable itself can sever the very feedback that constitutes the control. Stillness under a clamp doesn't show that nothing was being controlled.

For the purposes of this paper, corrective control has four evidential marks. The evidence should include a perturbation that threatens the projectible profile, a variable or relation to which the response pathway is dynamically coupled, a response that changes lower-scale conditions, and a result in which that response preserves a higher-scale relation supporting projection. Feedback on its own doesn't satisfy these conditions, because the evidence has to show that the feedback explains why the profile remains projectible under perturbation.

That evidence is a good deal more specific than the claim that a property cluster is stable, or that its properties stand in some causal order. It marks the most demanding diagnostic question: not just whether the profile persists, and not just whether it is ordered, but whether a specified corrective pathway preserves the relation that makes the profile projectible.

7 The diagnostic grid

The diagnostic grid is a set of evidential questions rather than an exhaustive taxonomy or a guaranteed sequence. Control-secured cases will usually display stability, network order, and maintenance as well, but the inference doesn't run in reverse. A stable cluster needn't have an internally ordered profile, a network-ordered kind needn't have a specified maintainer, and a maintained kind needn't be maintained by corrective control.

The grid also has off-diagonal cases. A bureaucratic status label may be maintained by forms and offices while its associated properties remain weakly ordered; a transient machine-failure profile may have ordered causal dependencies without any stabilizer that keeps it in place.

The grid isolates five questions that are recurrently compressed in HPC discussions: what gets projected, what profile supports the projection, what order makes the profile non-accidental, what maintains the profile, and whether that maintenance is corrective control. Selection, drift, institutional scaffolding, exemplar effects, and path dependence may each stabilize projectible profiles in ways that deserve separate treatment. These cases matter because they may secure projectibility without fitting neatly into the strict control role; the grid is meant to prevent conflation, not to classify every stabilizing process.

This is the constructive response to Onishi and Serpico's dilemma. Slater-style cliquish stability, Khalidi-style causal-network order, Boyd-style maintenance, and control-theoretic homeostasis are separated because no one of them should be made to carry all the weight of natural-kind realism.

Diagnostic demand	Evidence needed	Failure mode
Projection target	A specified field-relative inference, prediction, explanation, or intervention the category is supposed to support	Kind talk without saying what the category lets us project
Stable profile	A Slater-style projection from observed subcluster to unobserved cluster features or behavior across relevant contexts, plus invariance evidence for explanatory or interventional payoffs	Mere recurrence without projectible payoff
Network order	An ordering relation that explains which projections are licensed, such as core-to-derivative, conditioning, licensing, or causal-normative dependence	Causal or ordering vocabulary used for an unordered list or mere cue
Maintenance	A stabilizer whose removal would make the profile fragment, drift, or dissolve	Treating persistence as if a maintainer had been specified
Corrective control	Perturbation-sensitive correction that preserves a higher-scale relation through lower-scale variation	Treating stability, maintenance, or feedback as control

Table 1: A diagnostic grid for projectible profiles and homeostatic control.

The grid is meant to turn the question around. It asks the analyst to name the intended projection first and then justify each stronger securing claim, ending only where a maintainer corrects perturbations. The aim is to make the evidential demand behind HOMEOSTATIC explicit.

Nothing in the argument requires every historical use of HOMEOSTATIC in HPC theory to have meant feedback control. The next step is to test separability in cases: stable profile, network order, maintenance, and corrective control should answer different evidential questions and come apart in practice, not just in the table.

8 The cases

The cases are arranged by evidential role. Their taxonomic standing differs: the Watt governor calibrates the control architecture; glycemic regulation supplies a biological profile plausibly secured by such architecture; interjection tokens show projectibility and maintenance without a specified controller; and monitored agreement production shows what a linguistic dependency would have to add to count as control-secured in an online repair loop. Together, the cases test whether a profile supports projection, what order explains that projection, what maintains the profile, and whether the maintainer is corrective control.

The spread across machine, biological, and linguistic patterns is part of the test. The diagnostic concerns how projectibility is secured, a question not internal to any one science. If the same grid discriminates among the governor, glycemic regulation, interjection tokens, and monitored agreement production, the distinction belongs to the theory of kinds rather than the folklore of a particular field.

8.1 Calibration case: Watt governor

The Watt governor is a useful calibration case because its control structure is explicit. The governor-engine assembly is the system; flyballs, linkage, and valve relations are controller components; shaft speed is the lower-scale feedback variable; and the higher-scale controlled relation matches steam admission to workload. A change in engine speed moves the governor's flyballs; their position changes the valve aperture; the aperture changes the flow of steam; and the steam flow changes the speed of the shaft. A clamp placed directly on shaft speed would sever the incoming causal relations and destroy the control relation altogether (Weinberger, 2026).

Direct clamping still leaves ordinary perturbation tests available. A change in workload or demand is the right kind of perturbation, because it lets the feedback relation operate and reveals whether component-level adjustment preserves the higher-scale matching relation.

The governor shows why HOMEOSTATIC shouldn't mean constant. Its component-level variables are allowed to vary precisely so that steam admission matches workload across changes in speed. That pattern calibrates the strict control demand: a perturbation is met by responsive adjustment, and a higher-scale relation is preserved through that adjustment.

The governor's role is calibrational. As an artifact with a target fixed by its designer, it displays control-system predictability for comparison with Boyd-style kind projectibility. Candidate kind claims can then be tested by asking whether an analogous perturbation-sensitive architecture secures their projectible profiles. The governor doesn't calibrate everything that matters, since its target is fixed externally. It calibrates lower-scale feedback producing higher-scale matching; the next case supplies the harder lesson about how a range can be constituted from within the system.

8.2 Biological extension: glycemic regulation

Glycemic regulation is included because it shows why the control diagnostic should be relation-preserving rather than set-point-preserving. The relevant candidate is the normoglycemic regulatory profile of an organism, not glucose as a substance or a single blood value. The regulated variable is blood glucose concentration; the maintaining architecture is the organism's endocrine and metabolic organization; and the projectible profile consists in coordinated responses across feeding, fasting, activity, and other perturbations.

The standard feedback representation already gives the case a control-like architecture. It includes a device that sets a normal range, a sensor that measures the regulated variable, a detector that compares the measured value against the range and emits an error signal, a controller, and an effector that moves the variable back toward the range (Bich et al., 2020). Taken at face value, this is why the case looks control-secured.

The projectible payoff is the organism's metabolic response across changing conditions. Given the normoglycemic regulatory profile, one expects coordinated changes in insulin, glucagon, hepatic output, and glucose uptake across feeding, fasting, and activity, rather than an isolated reading that happens to fall within a range.

The standard representation already reserves the word CONTROLLER for the component that reads the error signal and drives the effector. The vocabulary of the calibration case and of the biological case coincides, and not by accident, since both are feedback architectures in which a sensed departure drives a correcting response. The diagram doesn't vindicate the word; the architecture has to earn it.

Two complications make that question sharper. The first concerns the set point, since population correlations for glucose and calcium may fit an equilibrium model better than they fit a defended set point (Fitzgerald & Bean, 2018).

The second is that Bich et al. (2020) argue that the feedback-loop diagram is explanatorily thin: it localizes functions too neatly, represents a hierarchical organism as a flat component chain, narrows the explanation to one regulated variable, and assumes a normal range without explaining what establishes it.

They move instead to organizational closure, on which glucose homeostasis is a manifestation of organismic self-maintenance and energy-resource management, not just the tracking of an externally given number. Read this way, the corrective structure is real, but the set point is the element the cybernetic analogy gets wrong.

That move marks the real contrast with the governor. The governor's target is set from outside by a designer, whereas the organism's range has to be explained from within the organized system. The higher-scale relation that glycemic control preserves is a relation between glucose availability and the organism's continued integrated functioning, held in place through wide variation in component-level variables across feeding, fasting, and activity. A fixed reading would misdescribe that relation. A biological case counts as control-secured only when the evidence displays this organized correction; mere range maintenance is insufficient.

Glycemic regulation is the biological stress test for the strict control diagnostic: the stable profile is maintained by organized responses to perturbation, and what's preserved is best understood as the relation rather than the reading. It also shows where the governor case stops: lower-scale feedback matters, but the biological target must be explained as part of organismic self-maintenance.

8.3 Contrast case: interjection tokens

English interjection tokens are the contrast linguistic case. The unit is token-level: a form counts here only when it occurs with the interjection profile of prosodic detachability, weak syntactic integration, interactional force, and uptake-relevance.[1] A detached response cry *shit* is an interjection token, while *shit* in an ordinary nominal environment is a noun token.

The case has two levels of projection. At the broad category level, an interjection token licenses

[1] For the broader claim that word-kinds can be understood as homeostatic property clusters, see Miller (2021). The present case is narrower: it asks which evidential demands interjection tokens and form-sequence constructions satisfy.

weak expectations about prosodic detachability, low syntactic integration, non-inflection in the relevant occurrence, interactional force, and uptake-relevance (Ameka, 1992; Norrick, 2009). Observing some of these properties licenses expectations about the others, which makes the profile projectible rather than a mere list.

Stronger projections arise at the lower constructional level. Forms such as *huh?*, *mmhm*, and *oh* support expectations about sequential position and interactional uptake, including repair initiation, continuation, and a change in epistemic state (Dingemanse, 2024), while forms such as *wow*, *oops*, *ouch*, *yuck*, *shit*, *hey*, *well*, and *uh* invite expectations about stance, event management, attention, transition, or delay. Norrick (2009) is helpful here because he treats interjections as an open-ended category with recurrent pragmatic-marker functions. The two-level structure matters because the diagnostic should not force projectibility to be all or nothing.

The contrast between *oh* and *uh* shows the kind of projection at issue. In an appropriate sequential position, *oh* leads us to expect a change-of-state uptake: the speaker has just registered, noticed, or reclassified something. *uh*, by contrast, leads us to expect delay or floor-holding rather than a completed uptake. These are genuine projections from form and placement to interactional role, but they don't yet show a controller preserving the use profile.

The list is heterogeneous, since several of these forms are analysed as discourse markers, fillers, summonses, or response tokens rather than as central interjections. This heterogeneity is part of the test, because the contrast case is a projectible profile of interjection tokens with kindhood partly at issue, rather than a classical kind with settled lexical boundaries.

The likely stabilizers include conventionalization, turn-taking routines, prosodic recognizability, acquisition, register, recurrent response patterns, and metapragmatic learning, and together these make the profile stable and useful. They also give parts of the profile some network order. The relevant order has to be asymmetric rather than mere non-independence: interactional autonomy conditions prosodic detachability and syntactic non-integration, sequential position conditions available uptake, and prosodic contour helps disambiguate the uptake at stake. That gives interjection tokens projectibility, some network order, and plausible stabilizers. It doesn't yet give them corrective control.

In this case, maintenance is best sought in recurrent interactional practices that keep form, prosody, sequential position, and expected uptake aligned across speakers. These practices operate at several levels. Conventionalization and community transmission entrench form–function pairings over time; acquisition, modelling, and metapragmatic teaching stabilize use across speakers; local repair handles occasional trouble in a conversation. Removing the recurrent practices wouldn't produce a single momentary error so much as drift in the projectible profile itself.

The homeostasis question is more demanding. For interjection tokens, the higher-scale relation would be a use-constructional relation linking turn position, prosodic packaging, syntactic detachment, and expected uptake. The lower-scale variables would include token form, intonation, duration, sequential placement, phonological reduction, and degree of syntactic detachment.

Given the token-level unit, relevant perturbations have to stay inside an otherwise interjectional occurrence. Ordinary category shifts, such as *wow* becoming an attributive modifier or *shit* appearing as an ordinary noun, move to a different construction. Better candidates are anomalous prosody, misplaced sequential deployment, over-integration in an otherwise interjection token, or an ambiguous token that receives repairable uptake. Failure of uptake is an outcome or a trigger for repair, rather than the perturbation itself.

A control claim would have to identify one loop rather than move among several possibilities. In an online interactional loop, failed or ambiguous uptake could lead to clarification or repair that restores sequential intelligibility. In a developmental loop, learner misuse could lead to correction or modelling and convergence on adult use. In a community-transmission loop, repeated uptake patterns could entrench a form–function pairing. These are neighbouring stabilizers, not yet one control architecture.

The contrast with monitored agreement production turns on evidential tractability, not on a special standard for interjections. The online interjection loop may use the same broad parsing-and-monitoring capacities used in speech repair. Agreement supplies a sharply specified relation, controller features to target forms, and an independently characterized perturbation. Interjection uptake offers a more diffuse relation and perturbation classes, such as anomalous prosody, misplaced sequencing, or over-integration, without an equivalent paradigm. Until the relevant loop is specified, interjection tokens remain stable, partly network ordered, or maintained without yet being control-secured.

8.4 Boundary case: monitored agreement production

Monitored agreement production is the boundary case where the homeostatic label comes closest to being earned. The candidate is the monitored production process preserving an utterance-level agreement relation, not the type-level grammatical regularity by itself.

In agreement systems, agreement is systematic covariance between controller and target: one expression fixes features that other expressions must express. *I go* is the canonical English pairing, while **I goes* mismatches the first-person controller with a third-person verb form. In Corbett's terms, the agreeing expressions are targets, and canonical agreement is the case in which the features take matching rather than mismatching values (Corbett, 2006). The features may be number, person, gender, class membership, or case. The case concerns monitoring across agreement features, with gender as one possible feature.

That terminology is useful but not decisive. Corbett's controller–target relation identifies a dependency structure in agreement. By itself, that relation doesn't amount to a feedback-control architecture. The controller doesn't sense a mismatch, compare target forms with a standard, or issue a corrective signal. Agreement morphology establishes the relation whose preservation matters; production monitoring and self-repair are where a control-theoretic claim has to look for feedback.

Agreement has a straightforward projectible payoff. Once the controller's features and the agreement domain are known, target forms can be projected across the domain. Number agreement lets us project verb form from subject number. Noun-class agreement lets us project class-marking across verbs, adjectives, and demonstratives (Ashton, 1947; Corbett, 2006). The dependency is also ordered rather than merely stable, because the controller–target relation is asymmetric: the controller supplies the feature values that agreement targets must match, even when the controller also bears or expresses those values.

For the grammatical dependency, the first evidential demands are straightforward: agreement has a projection target, a stable profile, and an ordered dependency relation. The harder question is whether any maintenance process for that relation is also corrective control.

This is where scale matters. Agreement as a type-level grammatical regularity may be stable and ordered, but online self-monitoring operates on particular utterances. Repairing *The key to the cabin-*

ets are to *The key to the cabinets is* preserves this utterance's agreement relation. The case is therefore a boundary case: token-level control evidence is possible without settling control of the broader grammatical kind.

The control question arises only when that dependency is embedded in a corrective process. Agreement attraction helps isolate one perturbation: plural local nouns reliably increase agreement errors, while animacy and length do not (Bock & Miller, 1991). The plural attractor perturbs the implementation of subject–verb number agreement; the agreement error is the failure state.

Levelt's repair model supports that loop only in a limited way. Speakers monitor inner or overt speech, interrupt, and repair (Levelt, 1983); this is parsing-based monitoring, not direct access to production machinery. The monitor is not agreement-specific and may also matter for interjection uptake. Agreement is cleaner because the monitored relation and perturbation can be specified independently: utterance-level agreement, target forms, and repair or recasting.

The agreement-specific character of the repair comes from the projected dependency itself. Once the controller's features and the domain are fixed, target forms are expected to express those values, so repair restores the higher-scale relation that made the profile projectible rather than only smoothing local fluency. The repair counts as agreement-specific only when the corrected form is selected because it restores the controller–target relation.

A simple attraction case makes the loop concrete:

(1) *The key to the cabinets are – sorry, is – on the table.*

The perturbing cue is the plural attractor *cabinets*; the threatened relation is agreement with *key*; the production-level variable is the verb form; and the response pathway is repair from *are* to *is*. The repair restores the dependency relation that licensed the projection from controller to target in the first place.

This keeps constraint and correction apart. A grammatical description may mark a mismatch as ill-formed without thereby supplying a corrective pathway. Corrective control requires the additional process by which a departure is detected and answered. The four marks may be met in the online repair loop: the mismatch threatens the projectible profile, agreement is the tracked relation, repair changes the target forms, and the higher-scale relation is preserved through that change.

Acquisition and normative correction are different candidate loops; coordination would be needed to make them parts of a single system. In acquisition, learners converge on agreement patterns over developmental time; in normative contexts, teachers, editors, or interlocutors may reject or correct mismatches. These processes help stabilize agreement and may provide further control-like evidence, but they operate at different scales from online monitoring. The honest conclusion is conditional: monitored agreement production supplies a specified linguistic loop, which avoids a loose analogy from grammatical dependency to feedback control, and the homeostatic label depends on that loop doing the relevant corrective work.

Grammaticality at large is a harder and more contested candidate.[2] If correction, repair, acquisition, transmission, and uptake together formed an architecture that returned usage to a community-sensitive profile, grammaticality could turn out to be control-secured, but whether grammaticality is a single kind, and which loops would be doing the controlling, remains in dispute.

[2] For the broader grammaticality-as-kind project, see Reynolds and Miller (2026). The present paper only uses monitored agreement production to show what a specified linguistic control loop would have to look like.

The diagnostic leaves the possibility open without granting it, and monitored agreement production shows how a control-secured linguistic case would have to specify its loop.

9 What the diagnostic changes for HPC theory

The grid matters for HPC-style applications in linguistics and social ontology. Linguistic categories can have projection targets, stable profiles, network order, and maintainers without all of them being homeostatic in the strict control-theoretic sense, and social practices can be stabilized by records, offices, sanctions, training, and uptake without thereby constituting single homeostatic control systems.

Social-status assignments are a clear case in point: they may support projection through causal-normative networks of uptake, recognition, transition, and constitutive ties without thereby carrying moral warrant (Reynolds, 2026). The diagnostic asks each case which evidential demands it satisfies before control is inferred from the mere fact of persistence.

A natural objection is that social correction and metalinguistic policing are themselves forms of feedback, and sometimes they do function that way. The concession strengthens the diagnostic because it means that some linguistic and social categories may indeed be control-secured, but only where the loop is specified at the right scale.

The monitored agreement case keeps that condition visible: its candidate controller is the monitored production loop preserving an utterance-level agreement relation. Broader categories would need their own coordinated architecture of correction, repair, acquisition, transmission, and uptake. The evidence has to earn the upgrade case by case; the label can't be granted on the strength of stability alone.

SPACER gives a useful proof of concept for this discipline. The dataset pairs naturalistic single-word substitution errors from Switchboard with speaker self-repairs and with comprehenders' offline text-editing corrections (Upadhye et al., 2025). The perturbation is a lexical substitution; the candidate tracked relations include lexical fit, semantic distance, phonemic distance, and contextual predictability; and the response pathways are speaker repair and comprehender replacement. That structure lets the diagnostic become an empirical comparison without assuming that all correction belongs to one loop.

Here SPACER matters methodologically. A dataset of this structure lets one ask whether the same perturbation measures predict speaker-side and comprehender-side correction, or whether apparent recovery of intelligibility is maintained by neighbouring stabilizers. Until the same tracked relation and response pathway are specified, speaker monitoring, comprehender editing, and intelligibility recovery shouldn't be folded into one homeostatic architecture.

This is also why target-declaration can't be gamed by relabelling. If speaker repair and comprehender editing show different relations to the perturbation measures, redescribing both as recovery of intelligibility doesn't make them one control loop. The measured contrast disciplines the target claim.

This constraint governs the upgrade from a declared projection target to a control claim. It doesn't choose the initial target from all possible descriptions of the profile; that choice still has to be justified by the projection, explanation, or intervention the category is meant to support.

The same distinction bears on the more formal side of HPC work. One temptation is to operationalize homeostasis as a return-to-basin claim, asking whether a variable comes back to a set point after perturbation. The governor and glycemic regulation both show why that test is too narrow, since what each system preserves is a matching relation: workload to steam flow in the one case, and glucose availability to self-maintenance in the other. In both, the relation is held in place through variation in the lower-scale variables.

A test that measures only the return of a variable to a fixed value can miss control that works precisely by allowing the variable to move. The basin is better defined over the relation the system preserves than over any set of fixed values.

For HPC-style work in linguistics this implies a division of labour. The home-field result is sober: negative or conditional, not a failed positive case. Interjection tokens show how far stable pragmatic projectibility can go without a specified controller, and monitored agreement production shows what a local control-secured loop can look like. Many categories that an HPC analysis might be tempted to call homeostatic are, on inspection, stable projectible profiles, maintained profiles, or network-ordered kinds, which are real and worth a realist's defence without being control-secured.

What philosophers of science gain is a domain-neutral test for those stronger claims. The diagnostic should also obey its own discipline: given a category already offered as a Boyd-style projectible kind, it specifies what additional evidence would be needed before each upgrade is licensed.

It's narrower than a general test for natural kindness, and it doesn't claim that all projectible categories share one stabilizing structure. This reflexive check is Boydian in spirit: Boyd treats the category NATURAL KIND itself as a natural kind in metaphysics and epistemology (Boyd, 1999, p. 168).

Its usefulness depends on whether it blocks overextension in cases like interjections while still recognizing control-secured cases, such as glycemic regulation or the monitored production loop that preserves an utterance-level agreement relation, when the relevant loop is specified.

10 Conclusion

Projectibility is the point throughout. The finding is separability: stable profile, network order, maintenance, and corrective control are distinct evidential achievements that can come apart. A stable cluster can underwrite projection without being control-secured, because the reliability of the profile for a specified inferential purpose is not yet evidence of corrective organization under perturbation. Reserving HOMEOSTATIC for corrective control is the recommendation that follows from that result, not the premise that produces it.

Data and code availability

The companion SPACER analysis is available online, and the source code is available in the project repository. The workflow downloads the public SPACER data from the project repository associated with Upadhye et al. (2025); the raw SPACER data are not redistributed here.

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